

Chapter 9

Cylindrical symmetry and coordinates

9.1 Charge near a grounded corner

Consider a point charge q near a corner formed by two grounded conducting planes meeting at an angle α , as depicted in the Figure. We want the potential ϕ inside the wedge. We use the image method: add mirror charges $\pm q$ until you close a circle. For example, for opening angles $\alpha_0 = 90^\circ, 60^\circ$,

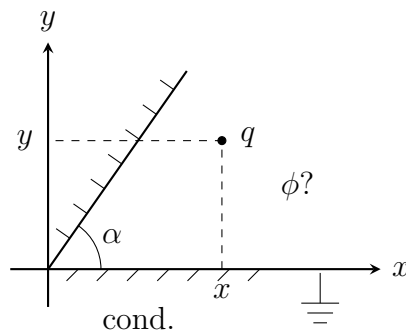


Figure 9.1: A point charge q near a corner formed by two grounded conducting planes meeting at an angle α ; we want the potential ϕ inside the wedge.

and 45° , the reflections close as shown in the Figure, with the image charges alternating in sign around a circle. In general

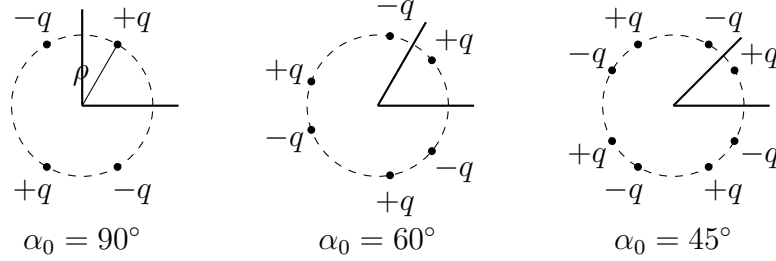


Figure 9.2: Image configurations for corners of opening angle $\alpha_0 = 90^\circ$, 60° , and 45° : mirror charges $\pm q$ are added, all at the same distance ρ from the corner, until the reflections close around a circle.

$$\alpha = \frac{\pi}{n} \quad (n \text{ even})[?] \Rightarrow 2n - 1 \text{ images} \quad (9.1)$$

Labeling the original charge and its images by the angles φ_i obtained from the reflections, all at the same distance r from the corner, the potential at a point (x, y, z) is

$$\phi = \sum_{i=1}^{2n} \frac{Q_i}{\sqrt{(r \cos \varphi_i - x)^2 + (r \sin \varphi_i - y)^2 + z^2}} \quad (9.2)$$

where

$$Q_i = (-1)^{i+1} q \quad (9.3)$$

9.2 Corner with no charge

Next, consider a corner with no charge inside: the two plates, meeting at an angle α , are held at potential ϕ_0 , as in the Figure, and we must solve $\nabla^2 \phi = 0$ in the wedge. (In the Green's function language we would need to find the Laplace piece F ; here we solve for ϕ directly.) We use separation of variables in cylindrical coordinates

$$\phi = R(r) F(\varphi) \quad (9.4)$$

with no z -dependence by symmetry. Substituting into the Laplace equation

$$\frac{r}{R} \frac{d}{dr} \left(r \frac{dR}{dr} \right) = \nu^2 = -\frac{1}{F} \frac{\partial^2 F}{\partial \varphi^2} \quad (9.5)$$

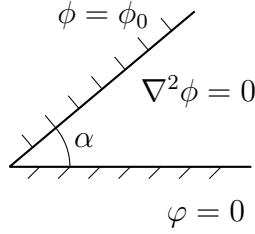


Figure 9.3: A corner of opening angle α with no charge inside; the conducting plates at $\varphi = 0$ and $\varphi = \alpha$ are held at potential ϕ_0 , and $\nabla^2\phi = 0$ in the wedge.

the angular equation being that of a simple harmonic oscillator. For the radial equation, try a power series

$$R(r) = \sum_{n=0}^{\infty} c_n r^n \Rightarrow n = \pm\nu \quad (9.6)$$

For $\nu \neq 0$, we then have

$$R(r) = a_\nu r^\nu + b_\nu r^{-\nu} \quad , \quad F(\varphi) = A \cos(\nu\varphi) + B \sin(\nu\varphi) \quad (9.7)$$

while for $\nu = 0$

$$R(r) = a_0 + b_0 \ln r \quad , \quad F(\varphi) = A_0 + B_0 \varphi \quad (9.8)$$

Since ϕ must be unique, $\nu \in \mathbb{Z}$ and $B_0 = 0$. The general solution is then

$$\begin{aligned} \phi(r, \varphi) = & a_0 + b_0 \ln r + \sum_{n=1}^{\infty} a_n r^n \sin(n\varphi + \alpha_n) \\ & + \sum_{n=1}^{\infty} b_n r^{-n} \sin(n\varphi + \beta_n) \end{aligned} \quad (9.9)$$

on the wedge $0 \leq \varphi \leq \alpha$. We want $\phi = \phi_0$ for all $r \geq 0$ at $\varphi = 0$ and $\varphi = \alpha$; together with regularity near $r \sim 0$, this requires

$$b_0 = 0 \quad , \quad A_0 a_0 = \phi_0 \quad , \quad A = 0 \quad , \quad b_\nu = 0 \quad (9.10)$$

and then

$$\phi(r, 0) = \phi(r, \alpha) \Rightarrow \nu = \frac{m\pi}{\alpha} \quad , \quad m = 1, 2, \dots \quad (9.11)$$

Hence

$$\phi(r, \varphi) = \phi_0 + \sum_{m=1}^{\infty} a_m r^{m\pi/\alpha} \sin\left(\frac{m\pi\varphi}{\alpha}\right) \quad (9.12)$$

The coefficients a_m are not fixed at this stage – one needs an additional boundary condition at large r , closing the domain. Near $r \sim 0$, the electric field and the surface charge density behave as

$$E \text{ and } \sigma \sim r^{\frac{\pi}{\alpha}-1} \quad (9.13)$$

so there is no charge in the corner.

9.3 Cylindrical Green's function

More generally, write

$$G = \frac{1}{|\vec{x} - \vec{x}'|} + F \quad (9.14)$$

where the first term is to be written in cylindrical coordinates, and F solves the Laplace equation in cylindrical coordinates. Separation of variables now takes the form

$$\phi = R(r) Q(\varphi) Z(z) \quad (9.15)$$

with R given by Bessel functions, Q by a simple harmonic oscillator, and Z by exponentials. One can show that

$$\begin{aligned} \frac{1}{|\vec{x} - \vec{x}'|} &= \frac{2}{\pi} \sum_{m=-\infty}^{\infty} \int_0^{\infty} dk e^{im(\varphi-\varphi')} \cos k(z-z') \\ &\quad \times I_m(kr_{<}) K_m(kr_{>}) \end{aligned} \quad (9.16)$$

where I_m, K_m are the modified Bessel functions.